Example 1 - Single Doc

R Holland

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1i. Summary

This rivt example calculates the maximum stress and deflection in a simply supported, uniformly loaded beam. It also serves as an annotated example of a single doc with multiple sections (a single doc does not use the report generating script).

The example illustrates the use of some of the most common API functions, commands and tags. Further details are provided in the rivt user manual.

The file may be formatted as a text, PDF or HTML doc by changing the type parameter in the PUBLISH command of the Doc API (rv.D) at the end of the file. Published files are found in the respective sub-folders of the _published folder.

2i. Load Combinations

This is an inline table using the restructured text syntax.

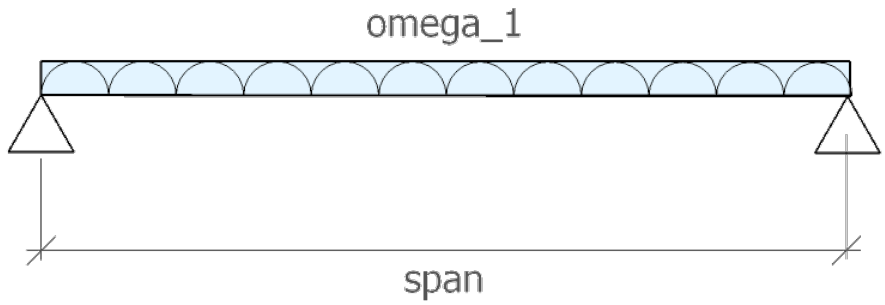
Table 1: ASCE 7-05 Load Effects

Equation No.	Load Combination
16-1	1.4(D+F)
16-2	1.2(D+F+T) + 1.6(L+H) + 0.5(Lr or S or R)
16-3	1.2(D+F+T) + 1.6(Lr or S or R) + (f1L or 0.8W)

When an inline table is contained in a TABLE block it produces the same output as above and also writes the table to a CSV file.

Table 2: ASCE 7-05 Load Effects (saved as csv in _stored folder)

Equation No.	Load Combination
16-1	1.4(D+F)
16-2	1.2(D+F+T) + 1.6(L+H) + 0.5(Lr or S or R)
16-3	1.2(D+F+T) + 1.6(Lr or S or R) + (f1L or 0.8W)



[Eq.1]

$$\sigma_1 = \frac{M_1}{S_1}$$

3v. Loads and Geometry

Successive lines of value definitions are formatted as a table. Variable values are defined with the define operator. The line tag [T] labels and numbers the table.

Table 3: Define Unit Loads

variable	value	[value]	description
D_1	3.80 psf	0.18 kPA	joists DL
D_2	2.10 psf	0.10 kPA	plywood DL
D_3	10.00 psf	0.48 kPA	partitions DL
D_4	3.00 klf	43.78 kN_m	fixed machinery DL
L_1	40.00 psf	1.92 kPA	ASCE7-O5 LL
b_1	10.00 inch	254.00 mm	beam width
h_1	18.00 inch	457.20 mm	beam depth
E_1	29000.00 ksi	199947.96 MPA	modulus of elasticity

The VALTABLE command reads variable values from the file in the _src folder. The text is used as the table title. The range specifies the starting and ending line to be read from the file (0:0 means all lines). The num;non parameter specifies whether theimported table is numbered.

Table 4: Beam Geometry [file: beam1.csv]

variable	value	[value]	description
spc_1	2.00 ft	0.61 m	beam spacing
spn_1	16.00 ft	4.88 m	beam span

Uniform Distributed Loads

Eq. 2 | dead load: ASCE7-05 2.3.2

```
dl_1 = 1.2*(D_4 + spc_1*(D_1 + D_2 + D_3))
```

D_2	spc_1	D_3	D_4	D_1
2.10 psf	2.00 ft	10.00 psf	3.00 klf	3.80 psf

dl_1	[dl_1]	reference
3.64 klf	53.09 kN_m	dead load: ASCE7-05 2.3.2

Eq. 3 | live load: ASCE7-05 2.3.2

```
ll_1 = 1.6*L_1*spc_1
```

spc_1	L_1
2.00 ft	40.00 psf

ll_1	[ll_1]	reference
0.13 klf	1.87 kN_m	live load: ASCE7-05 2.3.2

Eq. 4 | total load: ASCE7-05 2.3.2

```
omega_1 = dl_1 + ll_1
```

dl_1	ll_1
3.64 klf	128.00 ft·psf

omega_1	[omega_1]	reference
3.77 klf	54.96 kN_m	total load: ASCE7-05 2.3.2

4v. Beam Response

Table 5: Beam functions **[file:** src/sectprop.py **]**

Function	Docstring
rectsect(b, d)	section modulus of rectangle
rectinertia(b, d)	moment of inertia of rectangle
midspan_delta(ln, w, e, i)	mid-span deflection of simply supported beam with UDL

Eq. 5 | S-rectangle

```
section_1 = rectsect(b_1, h_1)
```

b_1	h_1
10.00 inch	18.00 inch

section_1	[section_1]	reference
540.00 in3	8849.01 cm3	S-rectangle

Eq. 6 | I-rectangle

```
inertia_1 = rectinertia(b_1, h_1)
```

b_1	h_1
10.0 inch	18.0 inch

inertia_1	[inertia_1]	reference
4860.0 in4	202288.5 cm4	I-rectangle

Bending Stress

Eq. 7 | mid-span UDL moment

```
                    2
      omega_1*spn_1
m_1 = --------------
            8
```

omega_1	spn_1
3.77 klf	16.00 ft

m_1	[m_1]	reference
120.52 ftkip	163.40 mkN	mid-span UDL moment

Eq. 8 | bending stress

```
          m_1
fb_1 = ---------
       section_1
```

m_1	section_1
120.5 ft2·klf	540.0 inch3

fb_1	[fb_1]	reference
2678.2 lb_in2	18.5 MPA	bending stress

Eq.9 | stress ratio

```
fb_1 < 20000*lb_in2
```

fb_1	20000*lb_in2	ratio	check	reference
2.68 ksi	20.00 ksi	0.133908	OK	stress ratio

Deflection

Eq. 10 | mid-span deflection

```
delta_1 = midspan_delta(spn_1, omega_1, E_1, inertia_1)
```

inertia_1	E_1	omega_1	spn_1
4860.00 inch4	29000.00 ksi	3.77 klf	16.00 ft

delta_1	[delta_1]	reference
0.04 inch	1.00 mm	mid-span deflection